

Coordination Compounds Master Handbook

Class 12 CBSE & JEE Advanced Special Edition

Chapter / Focus Area: Coordination Chemistry & Crystal Field Theory

1. Werner's Coordination Theory & IUPAC Nomenclature

- Primary Valency: Ionizable, corresponds to oxidation state of central metal ion.
 - Secondary Valency: Non-ionizable, corresponds to coordination number (geometry).
 - IUPAC Naming: Cation named first, anionic ligands end in -o (chlorido, cyanido).
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2. Valence Bond Theory vs Crystal Field Theory

- Octahedral Splitting (Δ_o): d-orbitals split into lower t_{2g} (d_{xy} , d_{yz} , d_{zx}) and higher e_g ($d_{x^2-y^2}$, d_{z^2}).
 - Spectrochemical Series: $I^- < Br^- < S^{2-} < SCN^- < Cl^- < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < EDTA^{4-} < NH_3 < en < CN^- < CO$
 - Strong field ligands (CN^- , CO) cause large Δ_o ($\Delta_o > P$), resulting in low-spin complexes (inner orbital d^2sp^3).
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3. Isomerism in Coordination Complexes

- Geometrical Isomerism: $[Pt(NH_3)_2Cl_2]$ has cis-platin (anti-tumor agent) and trans-platin.
 - Optical Isomerism: $[Co(en)_3]^{3+}$ exhibits d- and l- enantiomers.
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